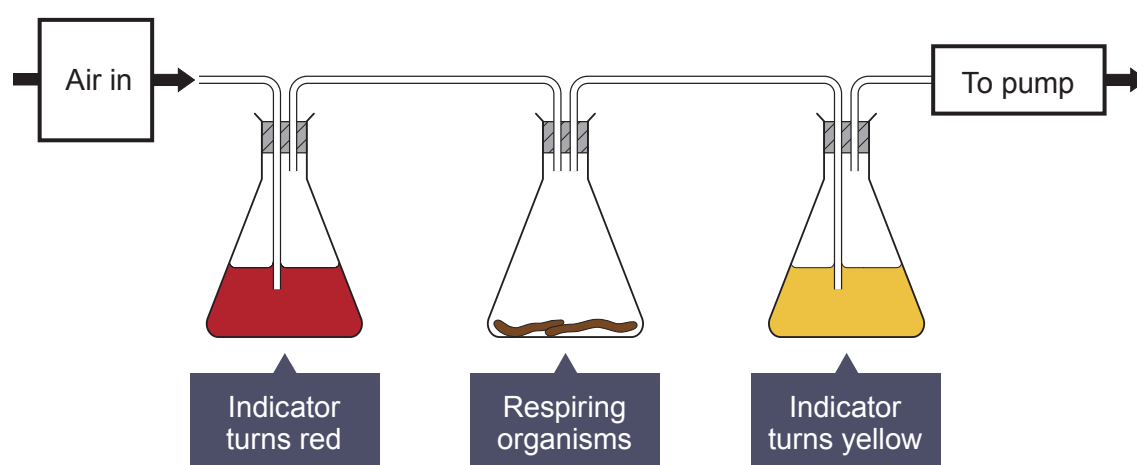


7. Hydrogen carbonate indicator can be used to measure the rate of respiration.

Concentration of carbon dioxide (%)	Colour of hydrogen carbonate indicator
less than 0.04	purple
0.04 (normal concentration in air)	red
greater than 0.04	yellow

The apparatus below can be used to compare the rate of respiration in small animals.



A group of students used different respiring organisms and recorded the time taken for the hydrogen carbonate indicator to turn yellow. The apparatus was kept at a constant temperature of 20 °C.

The results are shown in the table below.

Organism	Time taken for the hydrogen carbonate indicator to turn yellow (min)
earthworm	22
woodlouse	18
grasshopper	10
mouse	3

Examiner
only

(a) Explain why the hydrogen carbonate indicator changed colour in this experiment. [2]

.....

.....

.....

.....

(b) Give **two** reasons why the hydrogen carbonate indicator turned yellow fastest with the mouse. [2]

.....

.....

.....

.....

(c) The experiment was performed at a temperature of 20 °C. [2]

State **two** other variables that need to be controlled when comparing the respiration rates of different organisms.

.....

.....

(d) Describe how you would set up a control for this experiment and state the expected result. [3]

.....

.....

.....

.....

.....

.....

END OF PAPER

THE PERIODIC TABLE

Group

2
1

07

1 H Hydrogen 1																																				
7	Li	9	Be							11	B	12	C	14	N	16	O	19	F	20	Ne															
Lithium	3	Beryllium	4							Boron	5	Carbon	6	Nitrogen	7	Oxygen	8	Fluorine	9	Neon	10															
23	Na	24	Mg							27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar															
Sodium	11	Magnesium	12							Aluminium	13	Silicon	14	Phosphorus	15	Sulfur	16	Chlorine	17	Argon	18															
39	K	40	Ca	45	Sc	48	Ti	51	V	52	Cr	55	Mn	56	Fe	59	Co	59	Ni	63.5	Zn	65	Ga	70	Ge	73	As	75	Se	79	Br	80	Kr			
Potassium	19	Calcium	20	Scandium	21	Titanium	22	Vanadium	23	Chromium	24	Manganese	25	Iron	26	Cobalt	27	Nickel	28	Copper	29	Zinc	30	Gallium	31	Germanium	32	Arsenic	33	Selenium	34	Bromine	35	Krypton	36	
86	Rb	88	Sr	89	Y	91	Zr	93	Nb	96	Mo	99	Tc	101	Ru	103	Rh	106	Pd	108	Ag	112	Cd	115	In	119	Sn	122	Sb	128	Te	127	I	131	Xe	
Rubidium	37	Strontium	38	Yttrium	39	Zirconium	40	Niobium	41	Molybdenum	42	Technetium	43	Ruthenium	44	Rhodium	45	Palladium	46	Silver	47	Cadmium	48	Indium	49	Tin	50	Antimony	51	Tellurium	52	Iodine	53	Xenon	54	
133	Cs	137	Ba	139	La	179	Hf	181	Ta	184	W	186	Re	190	Os	192	Ir	195	Pt	197	Au	201	Hg	204	Pb	207	Bi	209	Po	210	At	210	Rn	222	Radon	86
Caesium	55	Barium	56	Lanthanum	57	Hafnium	72	Tantalum	73	Tungsten	74	Rhenium	75	Osmium	76	Iridium	77	Platinum	78	Gold	79	Mercury	80	Thallium	81	Lead	82	Bismuth	83	Polonium	84	Astatine	85			
223	Fr	226	Ra	227	Ac																															
Francium	87	Radium	88	Actinium	89																															
																					Key															

Key

relative atomic mass

Symbol	Name	atomic number
A_r		